TWO SUPERCONGRUENCES FOR CUBIC CENTRAL BINOMIAL SUMS

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Abstract. Let $A(N) = \sum_{k=0}^{N} \binom{2k}{k}^3$. We prove that every prime $p \equiv 3, 5, 6 \pmod 7$ satisfies $A(p^2) \equiv 8 + p^2 \pmod{p^3}$ and $A(p(p-1)) \equiv p^2 \pmod{p^3}$, resolving the first two conjectures of Peter Bala recorded in OEIS A079727. The main step is a polynomial identity for a second-order deformation of a truncated hypergeometric series. At every root away from 0 and 1, the identity determines a product of correction terms exactly. Combined with Zhi-Hong Sun's prime-level supercongruence, this gives $A(p^2-1) \equiv p^2 \pmod{p^3}$. A Vandermonde argument supplies the remaining endpoint. We also provide a single-file Lean formalization that proves the exact statements in Google DeepMind's Formal Conjectures repository, including a proof of the prime-level input and an explicit check of the axiom dependencies.

1. Introduction and main result

Write

$$c_k = \binom{2k}{k}, \qquad w_k = c_k^3, \qquad A(N) = \sum_{k=0}^{N} w_k \quad (N \geq 0). \tag{1}$$

The sequence $A(N)$ is entry A079727 in the On-Line Encyclopedia of Integer Sequences. In a comment dated July 12, 2024, Peter Bala proposed four congruence statements for this sequence [2]. We prove the first two. Their formal statements also appear in Google DeepMind's Formal Conjectures repository [1].

Theorem 1.1. *Let p be a prime with $p \equiv 3, 5, 6 \pmod 7$. Then*

$$A(p^2) \equiv 8 + p^2 \pmod{p^3}, \tag{2}$$

$$A(p(p-1)) \equiv p^2 \pmod{p^3}. \tag{3}$$

Moreover,

$$A(p^2 - 1) \equiv p^2 \pmod{p^3}. \tag{4}$$

The prime-level input is the following result of Zhi-Hong Sun.

Theorem 1.2 (Sun, [3, Theorem 3.2])**.** *If p is a prime with $p \equiv 3, 5, 6 \pmod 7$, then*

$$\sum_{k=0}^{(p-1)/2} w_k \equiv 0 \pmod{p^2}. \tag{5}$$

Passing from (5) to a sum of length p^2 introduces two correction terms. We control them by deforming the adjacent-coefficient recurrence for w_k and studying the resulting polynomial differential equations over $\mathbb{F}_p$. A conserved bilinear expression gives a polynomial with zero derivative. Its degree-p term is essential: the surviving polynomial is $8(1 - t^p)$, and its value at a root determines the second

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correction. This proves (4). The endpoint in (2) follows from Vandermonde's identity, while the tail removed in (3) vanishes by Lucas's congruence.

Throughout Sections 2–5, $p \geq 5$ is prime and $h = (p-1)/2$. Congruences between rational numbers are taken in $\mathbb{Z}_{(p)}$, the localization of $\mathbb{Z}$ at p. A bar denotes reduction modulo p when needed. We use $[s^j]F$ for the coefficient of s^j in F and $\theta = t\,d/dt$ for the Euler differential operator.

2. The correction coefficients

For $0 \leq k \leq h$, define $b_k, q_k \in \mathbb{Z}_{(p)}$ by

$$U_k(s) := 64^k \left(\frac{(1/2+s)_k}{(1+s)_k} \right)^3 \equiv w_k + s b_k + s^2 q_k \pmod{s^3}. \tag{6}$$

Here $(x)_k = x(x+1)\cdots(x+k-1)$, with $(x)_0 = 1$. The expression is expanded in $\mathbb{Z}_{(p)}[[s]]$; all constant terms of its denominators are units. In particular, $w_0 = 1$ and $b_0 = q_0 = 0$. The identity

$$(k+1+s)^3 U_{k+1}(s) = 8(2k+1+2s)^3 U_k(s)$$

gives the following three recurrences for $0 \leq k < h$:

$$(k+1)^3 w_{k+1} = 8(2k+1)^3 w_k, \tag{7}$$

$$(k+1)^3 b_{k+1} + 3(k+1)^2 w_{k+1} = 8(2k+1)^3 b_k + 48(2k+1)^2 w_k, \tag{8}$$

$$\begin{aligned}(k+1)^3 q_{k+1} + 3(k+1)^2 b_{k+1} + 3(k+1) w_{k+1} &= 8(2k+1)^3 q_k \\ &\quad + 48(2k+1)^2 b_k + 96(2k+1) w_k.\end{aligned} \tag{9}$$

These recurrences, together with the initial values, can equivalently be used as the definitions of the correction coefficients.

For comparison with harmonic-sum expansions, set

$$H_k = \sum_{j=1}^{k} \frac{1}{j}, \qquad H_k^{(2)} = \sum_{j=1}^{k} \frac{1}{j^2}, \qquad D_k = H_{2k} - H_k,$$

where all three expressions vanish at $k = 0$. Taking the first two logarithmic derivatives of (6) gives

$$b_k = 6 w_k D_k, \qquad q_k = w_k \big(18 D_k^2 - 6 H_{2k}^{(2)} + 3 H_k^{(2)} \big). \tag{10}$$

The recurrence formulation will be more convenient for the proof and for formal verification.

Lemma 2.1 (A translated block)**.** *For every integer $a \geq 0$ and $0 \leq r \leq h$,*

$$w_{ap+r} \equiv w_{ap} \big(w_r + ap\, b_r + a^2 p^2 q_r \big) \pmod{p^3}. \tag{11}$$

Proof. The exact adjacent-factor product for central binomial coefficients is

$$c_{ap+r} = c_{ap}\, 4^r \frac{(ap+1/2)_r}{(ap+1)_r}.$$

After cubing, the factor multiplying w_{ap} is $U_r(ap)$. Substituting $s = ap$ into (6) proves (11): the omitted terms are divisible by p^3 and every denominator remains a unit in $\mathbb{Z}_{(p)}$. □

3. A polynomial identity in characteristic p

Reduce the correction coefficients modulo p and define

$$f(t)=\sum_{k=0}^{h}\frac{w_k}{64^k}t^k,\qquad g(t)=\sum_{k=0}^{h}\frac{b_k}{64^k}t^k,\qquad q(t)=\sum_{k=0}^{h}\frac{q_k}{64^k}t^k\quad\text{in }\mathbb{F}_p[t]. \tag{12}$$

No residue restriction on p modulo 7 is imposed in this section.

Proposition 3.1 (Root identity)**.** *If $t_0\in\mathbb{F}_p\setminus\{0,1\}$ and $f(t_0)=0$, then*

$$f'(t_0)=g(t_0)=0,\qquad q(t_0)(\theta^2 f)(t_0)=1. \tag{13}$$

We prove the proposition by explicit polynomial identities. Introduce the differential operators

$$\begin{aligned}\mathcal{L}_0&=8\theta^3-t(2\theta+1)^3,\\ \mathcal{L}_1&=24\theta^2-6t(2\theta+1)^2,\\ \mathcal{L}_2&=24\theta-12t(2\theta+1).\end{aligned}$$

In these expressions, t multiplies the result of the operator to its right.

Lemma 3.2. *The polynomials in (12) satisfy*

$$\mathcal{L}_0 f=0,\qquad \mathcal{L}_0 g+\mathcal{L}_1 f=0,\qquad \mathcal{L}_0 q+\mathcal{L}_1 g+\mathcal{L}_2 f=0. \tag{14}$$

Moreover, $f(0)=1$ and $([t^h]f)^2=1$.

Proof. The coefficients of t^{k+1} in the three equations are (7)–(9), multiplied by $8/64^{k+1}$. Their constant coefficients are zero. The only additional coefficients, at degree $h+1$, vanish because $2h+1=0$ in $\mathbb{F}_p$. For the leading coefficient, $1/2=-h$ in $\mathbb{F}_p$, so

$$[t^h]f=\left(\frac{(1/2)_h}{h!}\right)^3=\left(\frac{(-h)_h}{h!}\right)^3=(-1)^h.$$

The assertion at $t=0$ follows directly from $w_0=1$. □

For polynomials u,v over a field of odd characteristic, put

$$\begin{aligned}\mathcal{B}(u,v)&=4t^2(1-t)(uv''+u''v-u'v')\\ &\quad+(4t-6t^2)(uv'+u'v)-tuv.\end{aligned} \tag{15}$$

For f,g,q as above, define

$$\mathcal{A}=\mathcal{B}(f,f), \tag{16}$$

$$\mathcal{K}=2\mathcal{B}(f,g)+8t(1-t)ff'-4tf^2, \tag{17}$$

$$\begin{aligned}\mathcal{J}&=2\mathcal{B}(f,q)+8(1-t)f^2+16t(1-t)fg'\\ &\quad-8t(1-t)f'g-4tfg.\end{aligned} \tag{18}$$

Lemma 3.3 (Conservation identities)**.** *For arbitrary polynomials f,g,q, the expressions above satisfy*

$$\theta\mathcal{A}=f\mathcal{L}_0 f, \tag{19}$$

$$\theta\mathcal{K}+2\mathcal{A}=f(\mathcal{L}_0 g+\mathcal{L}_1 f)+g\mathcal{L}_0 f, \tag{20}$$

$$\theta\mathcal{J}+\mathcal{K}=f(\mathcal{L}_0 q+\mathcal{L}_1 g+\mathcal{L}_2 f)+q\mathcal{L}_0 f. \tag{21}$$

Proof. Apply the product rule to (15)–(18) and collect the coefficients of the derivatives of f, g, q. For example,

$$\mathcal{L}_0 u = 8t^3(1-t)u''' + (24t^2 - 36t^3)u'' + (8t - 26t^2)u' - tu,$$

and the other two expanded operators are

$$\mathcal{L}_1 u = 24t^2(1-t)u'' + (24t - 48t^2)u' - 6tu, \qquad \mathcal{L}_2 u = 24t(1-t)u' - 12tu.$$

Substitution gives the three displayed identities, with no division by t or $1-t$. □

Lemma 3.4. *For the polynomials in* (12)*,*

$$\mathcal{A} = 0, \qquad \mathcal{K} = 0, \qquad \mathcal{J} = 8(1 - t^p). \tag{22}$$

Proof. Lemmas 3.2 and 3.3 imply

$$\theta\mathcal{A} = 0, \qquad \theta\mathcal{K} + 2\mathcal{A} = 0, \qquad \theta\mathcal{J} + \mathcal{K} = 0.$$

Write A_n, K_n, J_n for their coefficients at degree n. Then

$$nA_n = 0, \qquad nK_n + 2A_n = 0, \qquad nJ_n + K_n = 0.$$

If $n \neq 0$ in $\mathbb{F}_p$, the first two equations give $A_n = K_n = 0$. If $n = 0$ in $\mathbb{F}_p$, the second and third equations give the same conclusion. Thus $\mathcal{A} = \mathcal{K} = 0$ and $\mathcal{J}' = 0$.

Every term in (18) has degree at most $2h + 1 = p$. Consequently $\mathcal{J} = J_0 + J_p t^p$. Equation (18) gives $J_0 = 8f(0)^2 = 8$. To calculate J_p, let u, v, z be the coefficients of t^h in f, g, q, respectively; v and z may vanish. The coefficient of t^{2h+1} in (18) is

$$-2(2h+1)^2 uz - 4(2h+1)uv - 8u^2. \tag{23}$$

Indeed, $2\mathcal{B}(f, q)$ contributes the first term, the three terms involving g contribute the second, and $8(1-t)f^2$ contributes the third. Since $2h + 1 = 0$ and $u^2 = 1$ in $\mathbb{F}_p$, we obtain $J_p = -8$. □

Proof of Proposition 3.1. Let $R = t_0^2(1 - t_0)$, which is nonzero. Evaluating $\mathcal{A} = 0$ at a root of f gives $-4Rf'(t_0)^2 = 0$, hence $f'(t_0) = 0$. The other two identities in (22) now give

$$8Rf''(t_0)g(t_0) = 0, \qquad 8Rf''(t_0)q(t_0) = 8(1 - t_0^p) = 8(1 - t_0).$$

The second right-hand side is nonzero, so $f''(t_0) \neq 0$ and $g(t_0) = 0$. Cancelling $8(1 - t_0)$ gives $t_0^2 f''(t_0)q(t_0) = 1$. Finally, $\theta^2 f = t^2 f'' + tf'$, proving (13). □

Remark 3.5. The pairing (15) comes from the symmetric-square structure of the equation $\mathcal{L}_0 u = 0$. The coefficientwise argument above also explains why ordinary characteristic-zero reasoning about constants of differentiation is insufficient: the term t^p must be retained.

4. Summation over the blocks

Set

$$S = \sum_{k=0}^{h} w_k, \qquad B = \sum_{k=0}^{h} b_k, \qquad Q = \sum_{k=0}^{h} q_k, \qquad M_j = \sum_{k=0}^{h} k^j w_k \quad (j = 0, 1, 2), \tag{24}$$

with $k^0 = 1$, including at $k = 0$. Thus $M_0 = S$.

Corollary 4.1. *If $p \geq 5$ and $p \equiv 3, 5, 6 \pmod 7$, then*

$$M_1 \equiv B \equiv 0 \pmod p, \qquad QM_2 \equiv 1 \pmod p. \tag{25}$$

Proof. Sun's congruence gives $f(64) = S = 0$ in $\mathbb{F}_p$. Since $p \geq 5$ and $p \neq 7$, we have $64 \neq 0, 1$ in $\mathbb{F}_p$: the prime divisors of $64 \cdot 63$ are $2, 3, 7$. Apply Proposition 3.1 at $t_0 = 64$ and use

$$g(64) = B, \quad q(64) = Q, \quad (\theta f)(64) = M_1, \quad (\theta^2 f)(64) = M_2.$$

□

We use two consequences of Lucas's congruence. For every $a \geq 0$,

$$c_{ap} \equiv c_a \pmod{p}. \tag{26}$$

Also, if $h < r < p$, then

$$p \mid c_{ap+r}, \qquad w_{ap+r} \equiv 0 \pmod{p^3}. \tag{27}$$

For the latter statement, the units digits of $2(ap + r)$ and $ap + r$ are $2r - p$ and r, respectively, and $2r - p < r$. The corresponding binomial factor in Lucas's formula is zero.

Proposition 4.2. *If $p \geq 5$ and $p \equiv 3, 5, 6 \pmod 7$, then $A(p^2 - 1) \equiv p^2 \pmod{p^3}$.*

Proof. Define the outer moments

$$W_j = \sum_{a=0}^{p-1} a^j w_{ap} \qquad (j = 0, 1, 2).$$

Equations (26) and (27), applied also to $h < a < p$, imply

$$W_j \equiv M_j \pmod{p}. \tag{28}$$

By (27), only $0 \leq r \leq h$ contribute in each block. Lemma 2.1 therefore gives

$$\begin{aligned} A(p^2 - 1) &\equiv \sum_{a=0}^{p-1} w_{ap}\big(S + apB + a^2p^2Q\big) \\ &\equiv SW_0 + pBW_1 + p^2QW_2 \pmod{p^3}. \end{aligned} \tag{29}$$

Sun's theorem gives $S \equiv 0 \pmod{p^2}$, while $W_0 \equiv S \equiv 0 \pmod p$; hence $SW_0 \equiv 0 \pmod{p^3}$. Corollary 4.1 and (28) give $B \equiv W_1 \equiv 0 \pmod p$ and $QW_2 \equiv 1 \pmod p$. Substitution in (29) proves the result. □

5. The endpoint and the two conjectures

Lemma 5.1 (Vandermonde endpoint)**.** *For every prime $p \geq 5$,*

$$\binom{2p^2}{p^2} \equiv 2 \pmod{p^3}. \tag{30}$$

Proof. Vandermonde's identity gives

$$\binom{2p^2}{p^2} = \sum_{k=0}^{p^2} \binom{p^2}{k}^2. \tag{31}$$

If $1 \leq k < p^2$ and $p \nmid k$, the identity $k\binom{p^2}{k} = p^2\binom{p^2-1}{k-1}$ shows that $p^2 \mid \binom{p^2}{k}$. Its square contributes zero modulo p^3.

For $k = ap$, $1 \leq a < p$, cancel p in the same integer identity to get

$$a\binom{p^2}{ap} = p\binom{p^2 - 1}{ap - 1}.$$

The base-p digits of p^2-1 are $(p-1, p-1)$, while those of $ap-1$ are $(a-1, p-1)$. Lucas's congruence yields

$$\binom{p^2-1}{ap-1} \equiv \binom{p-1}{a-1} \equiv (-1)^{a-1} \pmod{p}.$$

Squaring and using that a is a unit in $\mathbb{Z}_{(p)}$, we obtain

$$\binom{p^2}{ap}^2 \equiv \frac{p^2}{a^2} \pmod{p^3}.$$

Inversion permutes $\mathbb{F}_p^\times$, and $\sum_{x\in\mathbb{F}_p^\times} x^2 = 0$ for $p \geq 5$. The latter follows by multiplying the summation variable by a nonzero element u with $u^2 \neq 1$; such an element exists since a quadratic polynomial has at most two roots. Thus $\sum_{a=1}^{p-1} a^{-2} \equiv 0 \pmod{p}$. The terms $k = 0, p^2$ in (31) each equal 1, so (30) follows. □

Proof of Theorem 1.1. For $p \geq 5$, Proposition 4.2 proves (4). Adding $c_{p^2}^3 \equiv 8 \pmod{p^3}$, by Lemma 5.1, proves (2).

For (3), consider $k = (p-1)p + r$ with $0 \leq r < p$. If $r > h$, (27) applies. If $r \leq h$, Lucas's congruence at the units digit gives

$$c_k \equiv \binom{2r}{r}\binom{2(p-1)}{p-1} \equiv 0 \pmod{p},$$

since the last factor has units digits $p-2 < p-1$. Hence $w_k \equiv 0 \pmod{p^3}$ for every $p(p-1) \leq k < p^2$. Removing the terms with $p(p-1) < k < p^2$ from (4) proves (3).

It remains to handle $p = 3$. The residues of w_k modulo 27 for $0 \leq k \leq 9$ are

$$1, 8, 0, 8, 19, 0, 0, 0, 0, 8.$$

Their sums through indices 6, 8, 9 are 9, 9, 17, respectively, which prove all three assertions at $p = 3$. □

6. Formal verification

The accompanying file `Peter Bala's Conjectures 1 and 2.lean` contains complete proofs of the two statements in `FormalConjectures/OEIS/79727.lean` at the repository revision specified in [1]. In the repository's notation, `OeisA3625.A p` means that p is prime and $p \bmod 7 \in \{3, 5, 6\}$, and `OeisA79727.a` is the sum A in (1). The proved declarations are

```
A79727Cubic.conjecture1
  A79727Cubic.conjecture2
```

with respective types

```
(p : ℕ) (hp : OeisA3625.A p) :
  OeisA79727.a (p ^ 2) ≡ 8 + p ^ 2 [MOD p ^ 3]

(p : ℕ) (hp : OeisA3625.A p) :
  OeisA79727.a (p * (p - 1)) ≡ p ^ 2 [MOD p ^ 3].
```

The file compares these types with the original declarations using Lean's definitional-equality checker. The source conjectures are read only for their types; their admitted proofs are not invoked.

The formalization follows the argument above, using (7)–(9) to define the correction coefficients. The names `cube_shift`, `full_invariants`, `field_moments`, `auxiliary`, and `endpoint_binomial` correspond to the translated-block formula, the three polynomial invariants, the moment identities, (4), and the Vandermonde endpoint. These declarations lie in the `A79727Cubic` namespace. The file also contains a proof of Sun's input as `A79727Proof.prime_case`; that proof uses finite polynomial identities and a quadratic-algebra lift.

Verification uses Lean 4.33.1 and mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`. All auxiliary proofs are included in one file, whose only imports are the Formal Conjectures statement module and two mathlib modules. Both final declarations depend only on `propext`, `Classical.choice`, and `Quot.sound`. The file contains no `sorry`, custom axiom, or `native_decide`, and an executable audit rejects any additional axiom dependency. In a project with these dependencies, it can be checked with

```
lake env lean "Peter Bala's Conjectures 1 and 2.lean"
```

The proof establishes the assertions uniformly for all primes in the stated residue classes. Finite computation is used only for the explicit case $p = 3$ in the proof.

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OpenAI's Codex was used extensively for mathematical exploration, proof development, Lean implementation, and the drafting and editing of this manuscript. This assistance is credited separately from the human authorship. The formal verification reported in Section 6 is the result of Lean's kernel checking the accompanying proofs and their axiom dependencies.

References

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[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A079727; conjectures contributed by Peter Bala, July 12, 2024. https://oeis.org/A079727.

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